

Kowdodi Phanendra Sree Harsh

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EDUCATION

- **National Institute of Technology, Rourkela** 2021-2025
B.Tech Degree in Mechanical engineering CGPA: 8.02/10.0
- **Urmi School, Vadodara** 2021
Class XII CBSE Percentage: 94.2%
- **Vrajbhoomi Int. School, Anand** 2019
Class X CBSE Percentage: 90.8%

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python

Operating Systems: Windows, Linux

Softwares/Frameworks: Arduino IDE, ROS1, Rviz, Gazebo, WeBots, , SolidWorks , Vrep (Coppeliasim)

Hardware: Jetson AGX Xavier, RaspberryPi, Arduino Uno, Esp32

Techstack: OpenCV, Numpy, Git, GitHub, MediaPipe

Coursework: Robotics, Data Structures, Algorithms, Computer Vision, OOPS, Control Systems, Neural Networks

Soft Skills: Leadership, Tough Situational Thinking, Management, Problem Solving

WORK EXPERIENCE

- **Accenture Solutions Private Limited** Oct 2025-Present
Advanced Application Engineer Analyst
 - Currently part of the Tools team, designing and developing **AI-powered agents** to support delivery teams in automating workflows.
 - Completed comprehensive training in Infrastructure Engineering and DevOps, with hands-on experience in **Java, Python, AWS, Docker, Kubernetes and Terraform**.
- **Real and Simulation Robot Research Lab, NIT Rourkela** July 2024-May 2025
Associate Researcher under [Prof Dayal R Parhi](#)
 - Contributed to multiple **underwater robot design** projects, focusing on electronic system development, sensor integration, and mechanical assembly.
 - Developed a low-computation **obstacle avoidance algorithm** for a mobile robot using Arduino and Raspberry Pi, optimized for resource-constrained environments.
 - Led a multidisciplinary team to prototype and test control systems for real-time navigation in uncertain environments.
- **Mechatronics Intern** May 2024-July 2024
Summer Intern at [Easy Lokal](#) [Certificate](#)
 - Engineered a functional **cooking robot** from the ground up, including the design and development of a detailed 3D CAD model with attention to modularity, space efficiency, and aesthetics.
 - Designed and implemented the complete electronic circuitry, integrating sensors, actuators, and a top-mounted camera for real-time process monitoring and control.
 - Conducted comprehensive analysis to ensure optimal performance and safety standards
- **eYantra Embedded systems and Robotics Lab, IIT Bombay** May 2023-July 2023
Summer Intern at [eYantra](#) [Video](#) | [Certificate](#)
 - Developed an **Autonomous Outdoor Navigation Stack (AONS)** using advanced sensors and the **Jetson AGX Xavier** for real-time, high-performance computation in outdoor robotics.
 - Integrated perception and planning algorithms for **3D depth, object detection, and mapping**.
 - Designed a complete **navigation pipeline** to enable reliable path planning and autonomous mobility in complex, unstructured outdoor terrains.
- **Robotics Trainer** 2018-2020
[World Robot Olympiad \(WRO\)](#)
 - Mentored secondary school students in robotics, programming, and analytical problem-solving as part of their preparation for the World Robot Olympiad.
 - Trained two teams that successfully advanced to **Nationals** for two years one after another.

PROJECTS

• MechanoTesla - The Desi Cybertruck

April 2024

Techstack: [ESP32](#), [Python](#), [SolidWorks](#), [HTML](#), [Web Socket](#)

[Repository](#) | [Video](#)

- Designed and **3D printed a Cybertruck-inspired mobile robot** with 4-mecanum wheels for omnidirectional movement.
- Integrated an ESP32 controller for real-time motor control and wireless navigation feedback.
- Built a **browser-based remote interface using HTML** and WebSockets for responsive robot control.

• Conveyor Sway Detector

Feb 2024

Techstack: [YOLO](#), [OpenCV](#), [Python](#)

- Designed a **real-time conveyor sway detection system** using an overhead camera and YOLO-based object detection to monitor belt misalignment in industrial settings.
- Collected and annotated a custom dataset from a **live steel plant environment**, ensuring robustness and real-world accuracy of the model.
- Enabled **early anomaly detection** to prevent potential equipment failure and downtime in automated material transport systems.

• Pixify - The OpenCV Game

May 2023

Techstack: [Python](#), [OpenCV](#)

[Repository](#)

- A new and **innovative image decoding challenge** to teach beginners the fundamentals of Python and OpenCV through hands-on problem-solving.
- Designed a **fully automated .exe-based evaluator** to validate user code, assess accuracy, and deliver instant feedback without manual intervention.

• PharmaBot - The Medicine Delivery Robot

Oct 2022 – Mar 2023

Techstack: [RaspberryPi](#), [Python](#), [OpenCV](#), [CoppeliaSim](#), [Path Planning](#)

[Repository](#)

- Developed an **autonomous robot** capable of **real-time line following using image processing** for navigation in a smart city simulation.
- Designed a **vision-based tracking system** using an overhead camera on the ceiling to monitor the bot in simulation for global positioning.
- Implemented **Dijkstra's Algorithm** with self tailored weights for path planning and to determine shortest delivery routes for efficient medicine distribution.

• Librarian Bot

Aug 2022 – Feb 2023

Tech Stack: [Python](#), [OpenCV](#), [ROS1](#), [SolidWorks](#), [Gazebo](#), [path planning](#)

[Video Demo](#)

- Implemented autonomous navigation system to locate and access designated bookshelves without manual control.
- Built a **Computer Vision pipeline** to detect barcodes and retrieve corresponding book data.

ACHIEVEMENTS

• ICORT 2025-4th IEEE INTERNATIONAL CONFERENCE ON RANGE TECHNOLOGY

March 2025

[Special Recognition](#)

[Credential](#)

- Co-designed and built an **innovative prototype system** to simulate sandstorm conditions for testing deep learning-based image restoration model.
- Represented NIT Rourkela at ICORT 2025, **DRDO ITR Chandipur** and received special recognition for an innovative and novel concept.

• eYantra Robotics Challenge, IIT Bombay

Apr 2023

[Second Runner-Up](#)

[Certificate](#)

- Secured **Rank 3 among 374 teams** for innovation and efficiency in autonomous delivery.
- Developed an autonomous medicine delivery robot using QR code detection and line following.
- Designed vision-based navigation with an onboard camera for accurate pickup and drop-off.

• Voyager, IIT Kharagpur

Jan 2023

[Second Runner-Up](#)

[Certificate](#)

- Ranked **3rd among the top 45 teams** in a national-level robotics competition.
- Developed a color-sensing line follower robot capable of autonomous navigation on colored paths.

• **Genius Olympiad 2017 - State University of New York, USA**

June 2017

Grand Gold Medalist

[Video](#) | [Medal](#) | [Govt. Recognition](#) | [Certificate](#)

- Secured **1st place among 53 participating countries** in the Robotics Category at the State University of New York, USA.
- Developed a robot to segregate waste blocks by color into designated bins.
- Designed the system with precise conveyor system and efficient actuation for real-time sorting.

• **FIRST LEGO LEAGUE 2016-17**

Jan 2017

Best Core Values Award

[Medal](#) | [Certificate](#)

- Awarded the team with the **Best Core Values Award** at FIRST LEGO League for innovation and professionalism.
- Designed and programmed a LEGO robot to complete precision tasks in the "**Animal Allies**" challenge.
- Showcased strong teamwork, creativity, and problem-solving under timed competition constraints.

• **World Robot Olympiad 2016**

Oct 2016

National Finalist

- **National finalist** for demonstrating innovative automation in environmental problem-solving.
- Built a robot to identify, classify, and **sort waste blocks by size and color** using sensors and control logic.

EXTRA CURRICULAR ACTIVITIES

President of Cyborg - Robotics and Automation Club of NIT Rourkela

May 2024-April 2025

[Website](#)

- Led a team of **86 as President of Cyborg**, the official Robotics Club of NIT Rourkela, driving projects in autonomous systems, computer vision, and embedded systems.
- Organized technical competitions and outreach events, enhancing club visibility, collaboration, and active member participation across campus.

Robotics Lead of Cyborg - Robotics and Automation Club of NIT Rourkela

May 2023-April 2024

- Led a **team of 25 in the software subsystem**, enhancing both my technical and managerial skills, while successfully delivering key project milestones under tight deadlines.

Event Manager, Death Race

Nov 2024

- Successfully organized and managed all rounds of a flagship robotics competition with **100 participating teams**, attracting over **3500 attendees** in two days, making it the most sought-after event of NIT Rourkela's annual techfest.

Event Manager, Robo Sumo

Nov 2023

- Led the successful execution of Robo Sumo, a major robotics competition at the annual techfest, with 25 participant teams and over **1500 attendees**.